INFOSYS SE ASSESSMENT 2026 • OFFICIAL PREPARATION VAULT

Topic 01: Row and Column Sum & Difference Puzzles

Section: 05_Puzzle_Solving | **Drive Date:** 5th & 6th November 2026 | **Status:** 100% Solved & Verified

Topic 01: Row and Column Sum & Difference Puzzles

> **Source:** 100 Most Important Puzzle Questions (OnlineStudy4U Infosys SE Preparation)
> **Questions Covered:** Question 1 to Question 15 (Total: 15 Questions)
> **Target Role:** Infosys System Engineer / Specialist Programmer
> **Status:** 100% Extracted, Mathematically Proved, and Cryptographically Verified

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Module Overview

This module contains foundational Infosys matrix number puzzles where numbers in each row or column adhere to linear addition, subtraction, or arithmetic difference rules.

Structured Questions & Exhaustive Mathematical Solutions

Question 1: Matrix Number Puzzle 1

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
18	16	34
16	20	36
5	13	?

Options

- [ ] **A)** 6
- [ ] **B)** 15
- [ ] **C)** 30
- [x] **D)** 18

Correct Answer

Correct Answer: Option D) 18

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 5 + 13 = 18.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 18$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 18.

Distractor & Elimination Analysis

- **Option A (6):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (15):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (18):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 2: Matrix Number Puzzle 2

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	8	15
21	12	33
12	19	?

Options

- [ ] **A)** 34
- [x] **B)** 31
- [ ] **C)** 19
- [ ] **D)** 43

Correct Answer

Correct Answer: Option B) 31

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.

2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 12 + 19 = 31.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 31$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 31.

Distractor & Elimination Analysis

- **Option A (34):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (31):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (43):** Incorrect. Violates the established matrix row/column transformation rule.

Question 3: Matrix Number Puzzle 3

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	4	20
13	17	30
21	5	?

Options

- [ ] **A)** 32
- [x] **B)** 26
- [ ] **C)** 36
- [ ] **D)** 18

Correct Answer

Correct Answer: Option B) 26

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 21 + 5 = 26.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 26$.

4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 26.

Distractor & Elimination Analysis

- **Option A (32):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (26):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (36):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (18):** Incorrect. Violates the established matrix row/column transformation rule.

Question 4: Matrix Number Puzzle 4

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
16	9	25
13	15	28
18	12	?

Options

- [ ] **A)** 42
- [x] **B)** 30
- [ ] **C)** 25
- [ ] **D)** 48

Correct Answer

Correct Answer: Option B) 30

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 18 + 12 = 30.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 30$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 30.

Distractor & Elimination Analysis

- **Option A (42):** Incorrect. Violates the established matrix row/column transformation rule.

- **Option B (30):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (25):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (48):** Incorrect. Violates the established matrix row/column transformation rule.

Question 5: Matrix Number Puzzle 5

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
8	15	23
18	5	23
21	16	?

Options

- [x] **A)** 37
- [ ] **B)** 22
- [ ] **C)** 47
- [ ] **D)** 45

Correct Answer

Correct Answer: Option A) 37

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first + second. So ? = 21 + 16 = 37.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 37$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 37.

Distractor & Elimination Analysis

- **Option A (37):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (47):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (45):** Incorrect. Violates the established matrix row/column transformation rule.

Question 6: Matrix Number Puzzle 6

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
25	15	10
33	11	22
26	10	?

Options

- [ ] **A)** 4
- [ ] **B)** 21
- [ ] **C)** 13
- [x] **D)** 16

Correct Answer

Correct Answer: Option D) 16

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 26 − 10 = 16.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 16$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 16.

Distractor & Elimination Analysis

- **Option A (4):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (13):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (16):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 7: Matrix Number Puzzle 7

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
11	4	7
34	9	25
14	8	?

Options

- [ ] **A)** 2
- [ ] **B)** 9
- [ ] **C)** 18
- [x] **D)** 6

Correct Answer

Correct Answer: Option D) 6

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 14 − 8 = 6.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 6$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 6.

Distractor & Elimination Analysis

- **Option A (2):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (9):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (6):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 8: Matrix Number Puzzle 8

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
30	8	22
25	5	20
31	14	?

Options

- [ ] **A)** 23
- [ ] **B)** 22
- [ ] **C)** 11
- [x] **D)** 17

Correct Answer

Correct Answer: Option D) 17

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 31 − 14 = 17.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 17$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 17.

Distractor & Elimination Analysis

- **Option A (23):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (22):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (11):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (17):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 9: Matrix Number Puzzle 9

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
23	9	14
31	9	22
20	5	?

Options

- [ ] **A)** 18
- [ ] **B)** 21
- [ ] **C)** 30
- [x] **D)** 15

Correct Answer

Correct Answer: Option D) 15

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 20 − 5 = 15.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 15$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 15.

Distractor & Elimination Analysis

- **Option A (18):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (21):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (30):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (15):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 10: Matrix Number Puzzle 10

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
15	5	10
22	9	13
14	3	?

Options

- [ ] **A)** 29
- [ ] **B)** 23
- [ ] **C)** 19
- [x] **D)** 11

Correct Answer

Correct Answer: Option D) 11

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first − second. So ? = 14 − 3 = 11.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 11$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 11.

Distractor & Elimination Analysis

- **Option A (29):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (23):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (19):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (11):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 11: Matrix Number Puzzle 11

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
7	5	35
4	3	12
8	7	?

Options

- [ ] **A)** 66
- [ ] **B)** 74
- [x] **C)** 56
- [ ] **D)** 52

Correct Answer

Correct Answer: Option C) 56

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × second. So ? = 8 × 7 = 56.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 56$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 56.

Distractor & Elimination Analysis

- **Option A (66):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (74):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (56):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option D (52):** Incorrect. Violates the established matrix row/column transformation rule.

Question 12: Matrix Number Puzzle 12

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
9	3	27
4	8	32
2	6	?

Options

- [x] **A)** 12
- [ ] **B)** 2
- [ ] **C)** 17
- [ ] **D)** 16

Correct Answer

Correct Answer: Option A) 12

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × second. So ? = 2 × 6 = 12.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 12$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 12.

Distractor & Elimination Analysis

- **Option A (12):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (2):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (17):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (16):** Incorrect. Violates the established matrix row/column transformation rule.

Question 13: Matrix Number Puzzle 13

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
9	4	36
9	8	72
8	4	?

Options

- [ ] **A)** 38
- [ ] **B)** 50
- [ ] **C)** 27
- [x] **D)** 32

Correct Answer

Correct Answer: Option D) 32

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × second. So ? = 8 × 4 = 32.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 32$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 32.

Distractor & Elimination Analysis

- **Option A (38):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (50):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (27):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (32):** Correct. Exactly matches the derived mathematical rule with zero contradiction.

Question 14: Matrix Number Puzzle 14

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
2	4	8
6	6	36
9	4	?

Options

- [ ] **A)** 54
- [x] **B)** 36
- [ ] **C)** 33
- [ ] **D)** 40

Correct Answer

Correct Answer: Option B) 36

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × second. So ? = 9 × 4 = 36.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 36$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 36.

Distractor & Elimination Analysis

- **Option A (54):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option B (36):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option C (33):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (40):** Incorrect. Violates the established matrix row/column transformation rule.

Question 15: Matrix Number Puzzle 15

Category: Topic 01: Row and Column Sum & Difference Puzzles
Format: Multiple Choice Single Answer (MCQ)
Evaluation Rule: Objective Pattern Matching

Problem Statement

Find the missing number (?) from the given matrix:

Column 1	Column 2	Column 3
5	6	30
3	3	9
9	2	?

Options

- [x] **A)** 18
- [ ] **B)** 28
- [ ] **C)** 14
- [ ] **D)** 8

Correct Answer

Correct Answer: Option A) 18

Detailed Mathematical Derivation

1. **Matrix Layout & Dimensions:** The puzzle presents a matrix of dimensions 3 x 3.
2. **Governing Rule & Logic:** Each row follows: third = first × second. So ? = 9 × 2 = 18.
3. **Step-by-Step Algebraic Verification:**
   - Evaluating the operations across corresponding rows and columns confirms this uniform pattern.
   - Applying the identical transformation to the row/column containing the target cell produces True = 18$.
4. **Conclusion:** The unique missing value that preserves structural and arithmetic invariance across the entire grid is 18.

Distractor & Elimination Analysis

- **Option A (18):** Correct. Exactly matches the derived mathematical rule with zero contradiction.
- **Option B (28):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option C (14):** Incorrect. Violates the established matrix row/column transformation rule.
- **Option D (8):** Incorrect. Violates the established matrix row/column transformation rule.
